

Alpaslan Kurt

Kocaeli, Türkiye

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Education

Koç University

M.Sc. in Industrial Engineering, GPA: 3.68/4.00

Istanbul, Türkiye

Expected: 2027

- Advisor: Sibel Salman Co-advisor: Ali Hassanzadeh
- Research focus: data-driven optimization and robust optimization

Gebze Technical University

B.Sc. in Industrial Engineering, GPA: 3.55/4.00 (High Honor)

Kocaeli, Türkiye

Graduated: 2025

Research Interests

Data-Driven Optimization, Robust Optimization, Decision-Focused Learning, Predict-then-Optimize, Optimization under Uncertainty, Operations Management

Research Experience

Graduate Researcher, Industrial Engineering Department

2025 – present

Koç University, Istanbul, Türkiye

Post-disaster telecommunication network recovery under uncertainty

Advisor: Sibel Salman Co-advisor: Ali Hassanzadeh (Alliance Manchester Business School)

- Optimizing the post-disaster deployment of Cells on Wheels (COWs) to restore mobile network coverage via a MILP that maximizes recovered demand net of deployment and dispatch costs, subject to COW and surviving base-station capacity, coverage, and depot dispatch/routing constraints.
- Capturing uncertainty in infrastructure damage, coverage, and demand through scenario-based stochastic, worst-case robust (max–min), and CVaR formulations, with a multi-period extension for dynamic COW relocation over time.
- Building an Istanbul (Avcılar) earthquake case study on Türk Telekom KPI and coverage data, prototyped in Gurobi/Python; manuscript in preparation.

Independent study (INDR 599): distributionally robust optimization for the newsvendor problem

Advisor: Fikri Karaesmen

- Studying distributionally robust optimization (DRO) for the data-driven newsvendor problem under demand distribution ambiguity.
- Employing Wasserstein and Kolmogorov–Smirnov ambiguity sets; reducing the DRO formulation to a regularized sample-average approximation (SAA) via strong duality.

Undergraduate Researcher, Industrial Engineering Department

2024 – 2025

Gebze Technical University, Kocaeli, Türkiye

GLORO: robust airfoil design optimization via a bilevel model-based methodology

Advisor: Figen Öztoprak Supported by TÜBİTAK 2209-B

- Developed GLORO, a bilevel methodology for constrained robust optimization with expensive (CFD-based) function evaluations, integrating Bayesian optimization with model-based derivative-free optimization (DFO).

- Approximated worst-case evaluations over the uncertainty set at the lower level using Gaussian Process surrogates and Bayesian optimization, while optimizing design variables at the upper level with local trust-region models robust to the resulting evaluation noise.
- Applied it to robust RAE 2822 airfoil shape design, maximizing the worst-case lift-to-drag ratio under Mach-number and angle-of-attack uncertainty with a minimum-thickness constraint.

Manuscripts

A. Kurt, D. Ş. Bük, Y. Yazıcıoğlu, F. Öztoprak, O. Son, G. Demir, “Robust Airfoil Design Optimization via a Bilevel Model-Based Methodology,” under review, Engineering Optimization.

Teaching Experience

Teaching Assistant, INDR 363 – Mathematical Programming Fall 2025, Spring 2026
Koç University

- Conducted weekly problem session classes and held office hours.
- Graded quizzes, homeworks, and exams.

Undergraduate Tutor, BENG 215 – Python for Bioengineers Fall 2023
Gebze Technical University

- Supported lab sessions and answered student questions on programming assignments.

Industry Experience

Production Management Intern, Hyundai Motor Türkiye Kocaeli, Türkiye
12/2024 – 07/2025

- Worked on short-term production planning and line balancing under capacity constraints.
- Worked on a car-sequencing algorithm to minimize color changes on the line.

Production Planning Intern, Turkish Aerospace Ankara, Türkiye
06/2024 – 07/2024

- Formulated and solved a CNC line-balancing problem via mathematical modeling in Python (PuLP).
- Supported production planning and part tracking for the TF-X (Turkish Fighter) program.

Material & Supply Chain Management Intern, Borusan Pipe Bursa, Türkiye
09/2024 – 12/2024

- Worked on MRP forecasting pipeline using machine learning, from data collection through model deployment.

Industrial Engineering Intern, Beyçelik Gestamp Kocaeli, Türkiye
08/2023 – 09/2023

- Built an integer program (Python, PuLP) to assign welding operations to machines under operational constraints.

Selected Projects

- **N-1 Secure Network Design** (INDR 584): Designed a reliability-aware capacitated facility-location model over Türkiye’s 81 provinces that keeps all demand satisfiable under any single hub failure. Formulated an expected-cost MILP with McCormick linearization and solved

it via multi-cut Benders decomposition (Gurobi), alongside greedy, swap, and simulated-annealing heuristics.

- **Predict-then-Optimize for Inventory and Lateral Transshipment** (INDR 522): Course project applying a data-driven predict-then-optimize framework to an inventory and lateral transshipment problem.
- **Airfoil Pressure Prediction**: Compared regression, random forest, SVR, and neural-network models for airfoil pressure prediction.

Skills, Languages, and Coursework

Technical Skills

- Non-Linear Programming; Robust Optimization; Surrogate Modeling; Machine Learning; Deep Learning
- Python, R, C, SQL, Git, Gurobi, PyEPO

Relevant Coursework

- Self-studied Stanford CS229 (Machine Learning); Deep Learning Specialization (in progress); Predict-then-Optimize (INDR 522)

Languages

- Turkish – Native English – Fluent

Honors and Awards

- KU Graduate Fellowship, awarded by Koç University (2025–2027)
- Graduated with High Honor, Gebze Technical University (2025)
- TÜBİTAK 2209-B research support (undergraduate research project)

References

Fikri Karaesmen, Professor of Industrial Engineering

Koç University

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Sibel Salman, Professor of Industrial Engineering

Koç University

Email: ssalman@ku.edu.tr

Ali Hassanzadeh, Assistant Professor of Management Science

Alliance Manchester Business School, University of Manchester

Email: ali.h@manchester.ac.uk

Figen Öztoprak, Assistant Professor of Industrial Engineering

Gebze Technical University

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